

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

**Practical - VI**

|                                |                                     |
|--------------------------------|-------------------------------------|
| Roll No. 43:                   | Name: NIKITA TIWARI                 |
| Class :SYIT                    | Batch : 1                           |
| Date of Assignment :29/07/2026 | Date/Time of Submission :29/07/2026 |

**AIM:** Tree Traversal: Write a program to:

- a. Implement pre-order,
- b. in-order,
- c. Post-order traversal of a binary tree.

CODE:

```
data structure 6.py - C:/Users/mvluc/Documents/data structure 6.py (3.14.6)
File Edit Format Run Options Window Help

class Node:
    def __init__(self,item):
        self.left=None
        self.right=None
        self.val=item
def inorder(root):
    if root:
        inorder(root.left)
        print(str(root.val)+"->",end='')
def postorder(root):
    if root:
        postorder(root.left)
        postorder(root.right)
        print(str(root.val) + "->", end="")
def preorder(root):
    if root:
        preorder(root.left)
        preorder(root.right)
        print(str(root.val) + "->", end="")
root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)
print("Inorder traversal")
inorder(root)
print("\npostorder traversal")
postorder(root)
print("\npreorder traversal")
preorder(root)
```

OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit
MD64] on win32
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/mvluc/Documents/data structure 6.py =====
Inorder traversal
4->2->1->
postorder traversal
4->5->2->3->1->
preorder traversal
4->5->2->3->1->
>> |
```

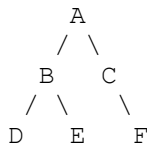
## Curiosity Questions:

### Instructions:

For each of the following binary trees, write the:

1. **Pre-order Traversal** (Root  $\rightarrow$  Left  $\rightarrow$  Right)
2. **In-order Traversal** (Left  $\rightarrow$  Root  $\rightarrow$  Right)
3. **Post-order Traversal** (Left  $\rightarrow$  Right  $\rightarrow$  Root)

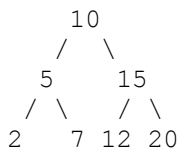
## Tree 1



### Answer:

- Pre-order:
- In-order:
- Post-order:

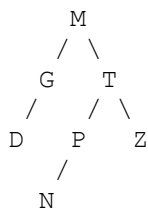
## Tree 2



### Answer:

- Pre-order:
- In-order:
- Post-order:

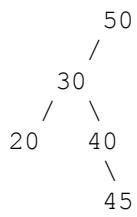
## Tree 3



### Answer:

- Pre-order:
- In-order:
- Post-order:

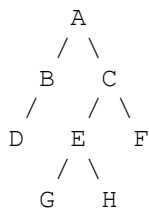
## Tree 4



**Answer:**

- Pre-order:
- In-order:
- Post-order:

## Tree 5



**Answer:**

- Pre-order:
- In-order:
- Post-order: